ELG 3125 - Signals & Systems

Lecture 2 - Part 2

Periodicity

Determine whether or not each of the following signals is periodic. If a signal is periodic, specify its fundamental period.

1.) $x(t) = e^{j100t}$

Condition for periodicity: $x(t) = x(t+T)$

$x(t+T) = e^{j(100)(t+T)} = e^{j100t}e^{j100T} = x(t) = e^{j100t}$

$\Rightarrow e^{j100t}e^{j100T} = e^{j100t}$

$\Rightarrow e^{j100T} = 1$

$\Rightarrow \cos(100T) + j\sin(100T) = 1 + j0$

$\Rightarrow \cos(100T) = 1$

$\Rightarrow 100T = 2\pi \Rightarrow T = \frac{2\pi}{100} \Rightarrow \boxed{T = \frac{\pi}{50}}$ $x(t)$ is periodic with period $T = \pi/50$.

2.) $x(t) = e^{(10+j100)t} = e^{10t}e^{j100t}$

e^{10t} not periodic $\Rightarrow x(t)$ not periodic

3.) $x(t) = e^{(10+j100t)} = e^{10}e^{j100t}$

$x(t+T) = e^{10}e^{j100(t+T)} = e^{10}e^{j100t}e^{j100T} = e^{10}e^{j100t}$

$\Rightarrow e^{10}e^{j100t}e^{j100T} = e^{10}e^{j100t} \Rightarrow e^{j100T} = 1 \Rightarrow \boxed{T = \frac{\pi}{50}}$ $x(t)$ is periodic with period $T = \pi/50$.

Periodic complex exponential and sinusoidal signals

$x(t) = e^{j\omega_0 t}$

Fundamental period: $T_0 = \frac{2\pi}{|\omega_0|}$

sinusoidal signal: $x(t) = A\cos(\omega_0 t + \varphi)$, $\omega_0 = 2\pi f_0$

Euler's relation: $\boxed{e^{j\theta} = \cos(\theta) + j\sin(\theta)}$ $\boxed{\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2}}$ $\boxed{\sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j}}$

Suppose we have: $x(t) = A\cos(\omega_0 t + \varphi)$

$x(t) = A\cos(\omega_0 t + \varphi) = A\left[\frac{e^{j(\omega_0 t+\varphi)} + e^{-j(\omega_0 t+\varphi)}}{2}\right] = \frac{A}{2}e^{j\varphi}e^{j\omega_0 t} + \frac{A}{2}e^{-j\varphi}e^{-j\omega_0 t}$

$\Rightarrow \boxed{A\cos(\omega_0 t + \varphi) = \frac{A}{2}e^{j\varphi}e^{j\omega_0 t} + \frac{A}{2}e^{-j\varphi}e^{-j\omega_0 t}}$

We know that: $e^{j\theta} = \cos(\theta) + j\sin(\theta)$, let $\theta = \omega_0 t + \varphi$ and multiply both sides by A, $A \in \mathbb{R}$

$\Rightarrow Ae^{j(\omega_0 t+\varphi)} = \underbrace{A\cos(\omega_0 t+\varphi)}_{\text{Re}} + j\underbrace{A\sin(\omega_0 t+\varphi)}_{\text{Im}}$ [a+jb]

$\Rightarrow A\cos(\omega_0 t+\varphi) = \mathrm{Re}\{Ae^{j(\omega_0 t+\varphi)}\} = A\,\mathrm{Re}\{e^{j(\omega_0 t+\varphi)}\}$

$\Rightarrow A\sin(\omega_0 t+\varphi) = \mathrm{Re}\{Ae^{j(\omega_0 t+\varphi)}\} = A\,\mathrm{Re}\{e^{j(\omega_0 t+\varphi)}\}$



In general if we have : $a+jb \Rightarrow |a+jb| = \sqrt{a^2+b^2}$

Suppose : $x(t) = e^{j\omega_0 t}$

$E_{period} = \int_0^{T_0} |x(t)|^2 dt = \int_0^{T_0} |e^{j\omega_0 t}|^2 dt$

$e^{j\omega_0 t} = \cos(\omega_0 t) + j\sin(\omega_0 t)$

$|e^{j\omega_0 t}| = |\cos(\omega_0 t) + j\sin(\omega_0 t)|$

$= \sqrt{\cos^2(\omega_0 t) + \sin^2(\omega_0 t)} = \sqrt{1} = 1$

$\boxed{|e^{j\omega_0 t}|^2 = 1}$

$E_{period} = \int_0^{T_0} dt = [t]_0^{T_0} = T_0 \Rightarrow \boxed{E_{period} = T_0}$

$P_{period} = \frac{1}{T_0}\int_0^{T_0} \underbrace{|e^{j\omega_0 t}|^2}_{=1} dt = \frac{1}{T_0}(T_0) = 1 \Rightarrow \boxed{P_{period} = 1}$

Euler's relation: $e^{j\theta} = \cos(\theta) + j\sin(\theta)$

$\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2}$

$\sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j}$